input · native OPG (2,808 × 1,316 px)



shared dispatcher - every image-consuming tool runs on base_image; the crop chain never compounds

input geometry: TPU = canonical views + static slots (no recompile); GPU = native so far - decision pending; tools always run native

every tool executes at full resolution

base_image

adapters

4

Policy

Qwen3.5-9B

reasoning node
thought + JSON tool calls

LoRA adapters:
SFT · GRPO (pluggable)

call

obs

Tool node



live crop

8 registered tools

zoom · window · denoise
enhance · contrast
locate · nudge
bilateral compare · fdi

continuous parameters:
strength · center · width

on answer / budget

final answer - findings JSON

```
{"findings":[{"quadrant":3,"tooth":34,"diagnosis":"Caries","confidence":0.82}]}
```

5 per-image findings

Evaluation

set-matched precision / recall / F1
over multi-finding predictions

95% bootstrap confidence intervals

heavy runs on GPU or TPU
(target decision pending - see input geometry)

probes (mechanism evidence):

- tool removal: with vs without
- perturbation: shown-box vs true-box
- nudge budget K sweep

(results pending, reported as open cells)

dev-split signal to training 6